



Published in final edited form as:

JAMA. 2026 February 24; 335(8): 663–664. doi:10.1001/jama.2025.26049.

Blood-Based Biomarkers for Alzheimer Disease:

Preventing Unintended Consequences

Heather E. Whitson, MD, MHS,

Nicole R. Fowler, PhD, MHSA,

Michelle M. Mielke, PhD

Duke Aging Center and Duke/UNC Alzheimer's Disease Research Center, School of Medicine, Duke University, Durham, North Carolina (Whitson); Geriatrics Research Education and Clinical Center, Durham VA Medical Center, Durham, North Carolina (Whitson); Department of Medicine, Indiana University School of Medicine, Indianapolis (Fowler); Indiana University Center for Aging Research at the Regenstrief Institute, Indianapolis (Fowler); Department of Epidemiology and Prevention, Wake Forest University School of Medicine, Winston-Salem, North Carolina (Mielke).

In May 2025, Fujirebio Diagnostics, Lumipulse G pTau 217/β-Amyloid 1-42 Plasma Ratio Test, a biomarker of Alzheimer disease (AD) pathology including amyloid plaques and tau tangles, became the first US Food and Drug Administration (FDA)–cleared blood test to help diagnose AD. In October 2025, Roche's Elecsys obtained FDA clearance for a plasma test of phosphorylated tau (pTau) 181 to rule out AD in primary care settings. Both tests are approved for individuals exhibiting cognitive decline. These developments increase access to a biologically based diagnosis of AD, which previously required lumbar puncture or positron emission tomography. A biologically based diagnosis determines eligibility for anti-amyloid therapies (eg, lecanemab, donanemab), which slow cognitive decline among those with early symptomatic AD. Although the risk-benefit ratio of available disease-modifying therapies for AD is modest and debated, the urgency for biomarker-based diagnosis will escalate if therapeutics with more compelling clinical benefit come to market. Patients not interested in or eligible for therapies may still want to learn their brain amyloid status to facilitate diagnosis and inform prognosis. Blood tests are cost-effective, minimally invasive, and highly accessible, which may help close the gap in AD diagnostic and care disparities, especially in primary care.

However, AD blood tests could result in unintended consequences and harm as they are expanded to real-world clinical populations. Individuals with multiple chronic conditions and societal hardships are most disadvantaged by poor access to cognitive assessments and criterion-standard diagnosis and may be at greatest risk of unintentional harm from new testing. Here we consider 3 areas of potential harm, and actions needed to mitigate risk, so that the blood tests achieve the intended goal of improving care for all.

Potential for Misdiagnosis

Although only 2 test platforms have received full FDA clearance, others are clinically available or in the pipeline. Patients and clinicians may not realize that there are multiple AD blood tests from which to choose or that their sensitivity and specificity vary.^{1,2} Further, positive and negative predictive values vary according to the patient's pretest probability of AD pathology. When a test with low to moderate specificity is administered to people with low pretest probability of brain amyloid (eg, patient with unimpaired cognition in primary care), a significant proportion of the positive results will be false-positive results. Likewise, administering a test with low to moderate sensitivity in patients with high pretest probability produces an unacceptable rate of false-negative results. As with any diagnostic test (eg, mammogram), people with false-positive results may be needlessly subjected to stress and/or risky or expensive treatments, whereas false-negative results deprive patients of potentially beneficial early treatment and care options. Inequities will widen if disadvantaged groups more often receive testing with less accurate, but perhaps more affordable, test platforms.

Most older adults with mild cognitive impairment and dementia have multiple chronic conditions and typically take at least 4 prescription medications.³ Specific chronic conditions, or their treatments, can impact the interpretation of the AD blood tests due to physiological reasons.⁴ For example, chronic kidney disease causes a significant increase in levels of AD blood proteins due to reduced kidney clearance. Sacubitril/valsartan, a medication for heart failure, inhibits an enzyme that alters the amyloid- β 42/40 ratio. Obesity can cause lower AD blood biomarker levels, relative to cerebral burden, due to higher blood volume and dilution of the proteins.⁵ Further, brain amyloid commonly co-occurs with Lewy body and cerebrovascular disease. Without multimodal testing and examination, overreliance on a single biomarker test will mislead diagnosis. The impact of comorbidities on diagnostic accuracy could exacerbate health disparities because of the higher prevalence of coexisting conditions among disadvantaged groups.

Uneven Access to the Next Step in the Care Pathway

The Lumipulse test utilizes 2 cut points, which means some portion of patients who are tested (usually 10%-20%) will receive results with intermediate values, which require follow-up with lumbar puncture or positron emission tomography to determine diagnosis. The Roche test has a high negative predictive value, but positive results require criterion-standard confirmation. If these criterion-standard tests are not available due to access or cost, patients may not benefit from the blood test. Likewise, if amyloid pathology is confirmed but the patient does not have access to or is unable to afford disease-modifying treatment, a critical inequity persists.

Legal and Psychological Harm

Even when test results are accurately interpreted and further testing and treatment options are available, testing can expose individuals and their families to legal or psychological harm. Although disclosure of biomarker results is well-tolerated by most, knowledge of test

results has been associated with depression, anxiety, and stress.⁶ Test results could impact employment, licensing decisions, access to long-term care facilities, or insurance eligibility. Medical records are discoverable, which means the information could be used in lawsuits. Because AD is a partially heritable disease, results could have implications for children and family members. Unless protections against these harms are mitigated and distributed equally in society, they will disproportionately impact those most vulnerable or least able to advocate for themselves.

Mitigate Risks and Unintended Consequences of Expanded Testing

Actions are needed to mitigate risks and unintended consequences of expanded testing. First, concerted education efforts are needed to inform multiple stakeholder groups about the changed landscape of AD diagnosis. Education for health care professionals should include information on the available AD blood tests and their limited performance data in specific care populations. Additionally, continuing medical education and medical boards must adjust curricula and content to enhance health care professionals' knowledge. Health systems and commercial laboratories must be equipped with information to select the most accurate test platforms and update as needed given the rapidly evolving field. Systematic reviews² and clinical practice guidelines⁷ are available and can be expanded and updated for primary care as evidence emerges. Laboratories should always report what assay platform was used and provide cut points. State medical societies can be instrumental in developing appropriate messaging and advocating for state-specific needs. Education materials are needed to prepare patients, lawmakers, insurers, and other stakeholders who may harbor a misunderstanding that positive AD biomarkers means that the patient has dementia. Patients need to be well-informed—before testing—about the potential benefits and risks of learning their results.

Second, advocacy groups must fight for appropriate legal protection for patients diagnosed with brain amyloid, including those with and without cognitive impairment. With blood-based testing, we expect AD pathology will be frequently detected in people without cognitive impairment, which poses unique legal and psychological ramifications. Although brain amyloid is a risk factor for future dementia, some individuals will remain unimpaired. Legal and ethical work is needed to minimize inappropriate bias against those who choose to know their biomarker status. Community and patient voices must be included. In a study including 554 primary care patients, two-thirds of the 63 who screened positive for dementia refused further diagnostic assessment, with reasons including fear of losing autonomy over living situation and driving.⁸

Third, test makers and researchers must collect and publish data about test accuracy in particular groups, such as those with comorbidities. Research examining AD blood tests in diverse populations is needed. Because race is a social construct, race-based differences in test performance, if they exist, likely reflect differences in social determinants of health. Therefore, race-specific cut points would be inappropriate and could widen diagnostic and treatment disparities.

Fourth, reimbursement models and policies must reduce barriers to criterion-standard testing and treatment for those who receive a blood test. This will be necessary to implement workflows that incorporate and incentivize biomarker testing—or an informed decision not to test—in the right patient at the right time. Because risks of harm are often greatest in the early days of disruptive medical progress, the time for action is now.

Funding/Support:

The authors' effort to draft and edit the submitted manuscript was supported, in part, by their roles in the Alzheimer's Disease Research Centers.

Conflict of Interest Disclosures:

Dr Whitson reported receiving grants from National Institutes of Health (5P30AG072958, R38AG065762, R21AG089947) and personal fees from American Geriatrics Society, Wake Forest University, DrivenData Inc, Yale University, and UpToDate and serving as co-chair of the Alzheimer's Association Clinical Practice Guideline on Blood-Based Biomarkers for Alzheimer's Disease. Dr Fowler reported receiving grants from the National Institutes of Health (U24AG082930) and serving on an advisory board for Roche Diagnostics.

Dr Mielke reported serving on an advisory board for Biogen, Eisai, Lilly, Neurogen Biomarking, Siemens Healthineers, Roche, and Novo Nordisk.

Role of the Funder/Sponsor:

The Alzheimer's Disease Research Centers had no role in the preparation, review, or approval of the manuscript or decision to submit the manuscript for publication.

REFERENCES

1. Teunissen CE, Kolster R, Triana-Baltzer G, Janelidze S, Zetterberg H, Kolb HC. Plasma p-tau immunoassays in clinical research for Alzheimer's disease. *Alzheimers Dement*. 2025;21(1):e14397. doi:10.1002/alz.14397 [PubMed: 39625101]
2. Pahlke S, Kahale LA, Mahinrad S, et al. Blood-based biomarkers for detecting Alzheimer's disease pathology in cognitively impaired individuals within specialized care settings: a systematic review and meta-analysis. *Alzheimers Dement*. 2025;21(11):e70828. doi:10.1002/alz.70828 [PubMed: 41193403]
3. Schubert CC, Boustani M, Callahan CM, et al. Comorbidity profile of dementia patients in primary care: are they sicker? *J Am Geriatr Soc*. 2006;54(1): 104–109. doi:10.1111/j.1532-5415.2005.00543.x [PubMed: 16420205]
4. Mielke MM, Fowler NR. Alzheimer disease blood biomarkers: considerations for population-level use. *Nat Rev Neurol*. 2024;20(8):495–504. doi:10.1038/s41582-024-00989-1 [PubMed: 38862788]
5. Mohammadi S, Rahmani F, Dolatshahi M, Schindler SE, Raji CA; ADNI Collaborators. Effects of obesity on plasma biomarker and amyloid PET trajectories in Alzheimer's disease. *Alzheimers Dement (Amst)*. 2025;17(3):e70143. doi:10.1002/dad2.70143 [PubMed: 40620424]
6. van der Schaar J, Visser LNC, Bouwman FH, et al. Considerations regarding a diagnosis of Alzheimer's disease before dementia: a systematic review. *Alzheimers Res Ther*. 2022;14(1):31. doi:10.1186/s13195-022-00971-3 [PubMed: 35144684]
7. Palmqvist S, Whitson HE, Allen LA, et al. Alzheimer's Association Clinical Practice Guideline on the use of blood-based biomarkers in the diagnostic workup of suspected Alzheimer's disease within specialized care settings. *Alzheimers Dement*. 2025;21(7):e70535. doi:10.1002/alz.70535 [PubMed: 40729527]
8. Fowler NR, Frame A, Perkins AJ, et al. Traits of patients who screen positive for dementia and refuse diagnostic assessment. *Alzheimers Dement (Amst)*. 2015;1(2):236–241. doi:10.1016/j.dadm.2015.01.002 [PubMed: 26258162]